



Cross-species extrapolation of chronic nickel Biotic Ligand Models

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ABSTRACT

The use of Biotic Ligand Models (BLMs) to normalize metal ecotoxicity data and predict effects in non-BLM organisms should be supported by quantitative evidence. This study determined the ability of chronic nickel BLMs developed for the cladocera *Daphnia magna* and *Ceriodaphnia dubia* to predict chronic nickel toxicity to three invertebrates for which no specific BLMs were developed. Those invertebrates were the snail *Lymnaea stagnalis*, the insect *Chironomus tentans*, and the rotifer *Brachionus calyciflorus*. Similarly, we also determined the ability of chronic nickel BLMs developed for the alga *Pseudokirchneriella subcapitata* and the terrestrial vascular plant *Hordeum vulgare* to predict chronic nickel toxicity to the aquatic vascular plant *Lemna minor*. Chronic nickel toxicity to the three invertebrates and the aquatic plant were measured in five natural waters that varied in pH, Ca, Mg, and dissolved organic carbon (DOC), which are known to affect chronic nickel toxicity and are the important input variables for the chronic nickel BLMs. Nickel toxicity to the three invertebrates varied considerably among the test waters, i.e., a 14-fold variation of EC50s in *L. stagnalis*, a 3-fold variation in EC20s in *C. tentans*, and a 10-fold variation in EC20s in *B. calyciflorus*, but the cladoceran BLMs were able to predict nickel effect concentrations within a factor of two. Nickel toxicity (EC50s) to *L. minor* varied by 6-fold among the test waters. Although the *P. subcapitata* and *H. vulgare* BLMs offered reasonable predictions of nickel EC50s to *L. minor*, the *D. magna* and *C. dubia* BLM showed better predictions. Our results confirm the influence of site-specific pH, hardness, and DOC on chronic nickel toxicity to aquatic organisms, and support the use of chronic nickel BLMs to manage this influence through normalizations of ecotoxicity data.

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1. Introduction

The influence of abiotic factors (e.g., hardness, pH, dissolved organic matter, etc.) on the toxicity of metals to aquatic organisms is increasingly being incorporated into ecological risk assessments and the derivation of water quality criteria for metals (ECB, 2003; U.S. EPA, 2007), and has prompted the development of bioavailability models to predict acute and chronic metal toxicity (Santore et al., 2001; De Schamphelaere and Janssen, 2002; De Schamphelaere et al., 2002). As such, chronic freshwater bioavailability models for nickel (Ni) have been developed for artificial test media and validated with field waters for several organisms across different trophic levels, including a unicellular alga (*Pseudokirchneriella subcapitata*) (Deleebeek et al., 2009a,b), two invertebrates (*Ceriodaphnia dubia* and *Daphnia magna*) (De Schamphelaere et al., 2006a; Deleebeek et al., 2008), and a fish species (*Oncorhynchus mykiss*) (Deleebeek et al., 2007a). Additionally,

Ni BLMs have also been developed for the terrestrial vascular plant *Hordeum vulgare* based on hydroponic exposures (Lock et al. 2007; Antunes and Kreager, 2009). In general, toxicity of the free Ni^{2+} ion to each of these species is reduced with increasing concentrations of H^{+} ions (decreasing pH) and with increasing concentrations of hardness ions (Ca^{2+} and Mg^{2+}). Furthermore, binding of Ni^{2+} ions to dissolved organic carbon (DOC) (Van Laer et al., 2006) results in a decreased bioavailability of the dissolved Ni fraction (Deleebeek et al., 2008). These relationships result in situations where nickel toxicity can vary considerably among waters with different physico-chemical properties. For example, chronic nickel EC10s (concentration causing 10% effect in test population) varied by 8-fold for *D. magna* depending upon the source of the natural test water (Deleebeek et al. 2008).

The primary role of BLMs within chemicals management frameworks of metals is to remove the influence of test-specific abiotic conditions within ecotoxicity databases. Influences of abiotic factors may introduce variability and/or bias and therefore confound inter-species sensitivity comparisons, which are needed in the development and analysis of Species Sensitivity Distributions (SSDs) and, subsequently, environmental quality standards. Removing the influence of abiotic conditions allows for the normalization of individual data within

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the ecotoxicity database to a relevant combination of abiotic factors (Van Sprang et al., 2009), e.g., site-specific pH, hardness, and DOC in the case of nickel. For normalization to proceed, it is implicitly assumed that BLMs can be extrapolated within taxonomically similar groups, i.e., that BLMs developed for *O. mykiss* can be applied to ecotoxicity data for other fish species, that BLMs for *D. magna* and *C. dubia* can be applied to ecotoxicity data for other invertebrates, and that BLMs for *P. subcapitata* can be applied to ecotoxicity data for other algae. The validity of extrapolating BLMs is particularly important for substances like Ni where the ecotoxicity databases are taxonomically diverse, because physiological differences among the organisms in the database may affect mechanisms of toxicity and the way in which water quality variables like pH, Ca, Mg, and DOC may influence Ni toxicity.

An ecotoxicity database comprised of chronic Ni toxicity data for 31 freshwater aquatic species was accepted for regulatory risk assessment application in Europe (ECB, 2008). Support for cross-species extrapolation has been shown for applying the *O. mykiss* BLM to another fish, i.e. *Pimephales promelas* (Deleebeek et al., 2007a), for applying the *D. magna* BLM to other cladocera (Deleebeek et al., 2007b) and for applying the *P. subcapitata* BLM to other green algae species (Deleebeek et al., 2009b). The results of the latter toxicity study also agreed well with uptake affinity constants for Mg and Ni in *Clamydomonas reinhardtii* (Worms and Wilkinson, 2007). However, no data existed to support the extrapolation of the cladoceran BLMs to other invertebrate species in the Ni toxicity database. Invertebrates have been shown to be among the most sensitive aquatic species to Ni exposure and are the most diverse taxonomic group in the Ni ecotoxicity database (ECB, 2008), with cladocera (De Schamphelaere et al., 2006a; Deleebeek et al., 2007b, 2008), insects (Nebeker et al., 1984), molluscs (Nebeker et al., 1986), and hydroids (Santiago-Fandino, 1983) being represented. The vascular plant *Lemna minor* is among the most sensitive species to Ni exposure under conditions of relatively low pH and low DOC (ECB, 2008), but the ability of chronic Ni BLMs to predict Ni toxicity to aquatic vascular plants has not been verified.

The basis for a cross-species extrapolation is the assumption that the parameters which describe interactions between cations (notably Ca^{2+} , Mg^{2+} , and H^+) and the toxic free metal ion (e.g., Ni^{2+}) are constant across organisms, and that only intrinsic sensitivity varies among species (Di Toro et al., 2001; Van Sprang et al., 2009). This assumption would be supported by quantitative evidence showing that the available chronic Ni BLMs can predict toxicity for taxonomically dissimilar species with reasonable accuracy. As such, the purpose of this study was to investigate whether the existing BLMs for Ni can predict, with sufficient certainty, chronic toxicity of four aquatic species for which no specific BLM has been developed.

The validity of cross-species extrapolation was evaluated by determining: 1) the accuracy of BLM predictions for Ni toxicity; and, 2) the extent to which normalization can reduce intra-species variability for non-BLM species.

The four non-BLM species used to evaluate chronic Ni toxicity included three invertebrate species (the insect *Chironomus tentans* (now known as *Chironomus dilutus*), the fresh water snail *Lymnaea stagnalis*, and the rotifer *Brachionus calyciflorus*) and one vascular plant species (*L. minor*). These organisms were exposed to nickel in five surface waters that were expected to offer a range of low to high protection against nickel toxicity, based on their chemical properties.

2. Materials and methods

2.1. Surface water

Water was collected from five locations in the Western U.S. from sites selected to maximize variability in pH, hardness, and DOC (the major factors affecting Ni bioavailability and toxicity to organisms) (Table 1). Ranges of pH, hardness, and DOC for the five exposure waters were kept within the boundaries of these parameters used in

the development of the BLMs. The collection locations were identified from historical data retrieved from U.S. Geological Survey real-time water quality stations (<http://cfpub.epa.gov/surf/locate/map2.cfm>).

Water was collected in pre-rinsed 20-L polypropylene containers using a submersible pump and polypropylene tubing. Samples were immediately placed into coolers, transported to the testing laboratory and kept at $4 \pm 2^\circ\text{C}$ in the dark until testing began. Suspended solids concentrations ($<3\text{ mg/L}$) and indigenous macro-organisms (none observed visually) in water samples were negligible so water samples were not filtered prior to testing.

Water from two sites (Zollner Creek and pH-amended S. Platte River) was manipulated at the testing laboratory to achieve the desired water quality parameters necessary to maximize variability in bioavailability. Zollner Creek was adjusted from 32 mg/L as CaCO_3 hardness to 125 mg/L as CaCO_3 hardness by adding reagent grade chloride salts of calcium (CaCl_2 ; Fisher Scientific, Pittsburg, PA, USA) and magnesium (MgCl_2 ; Fisher Scientific, Pittsburg, PA, USA) while maintaining the molar ratio of these cations (Table 1). The South Platte River water was altered by acidification to pH 7.0 using dilute HCl (Fisher Scientific, Pittsburg, PA, USA) and placing test chambers into a sealed glass box that was injected with an appropriate amount of CO_2 gas to maintain a 3–7% CO_2 atmosphere, depending on the water surface area to volume ratio (Table 1) (Mount and Mount, 1992). Specifically, the average CO_2 atmosphere used for *L. stagnalis*, *C. tentans* and *B. calyciflorus* was 4.2%, 3.1% and 6.7% respectively.

2.2. Toxicity test methods

Toxicity testing with invertebrates was conducted by Parametrix Inc., Corvallis, OR (USA) and toxicity testing with *L. minor* was conducted by AquaTox Testing & Consulting Inc., Guelph, ON (Canada).

2.2.1. *C. tentans*

The toxicity study with *C. tentans* was conducted based on the OECD 219 standard test method for a 10-day exposure (OECD, 2006). Second instar larvae were obtained from Aquatic Biosystems (Fort Collins, CO). Larvae were held in an environmental chamber ($25 \pm 2^\circ\text{C}$) under light aeration and observed for 48 h prior to test initiation.

Exposure treatments were prepared by adding appropriate volumes of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ stock (J.T. Baker Chemical Co., Phillipsburg, NJ, USA) to site waters in polypropylene dilution chambers. Test waters were allowed to equilibrate for at least 18 h at test temperature prior to distributing to exposure chambers for test initiation or renewal. Nominal test concentrations ranged from 50 to 6400 $\mu\text{g Ni/L}$ and varied among waters as a function of Ni bioavailability (details available in Supplemental section). Exposures were initiated by randomly distributing organisms directly into test chambers. Test chambers contained approximately 50 ml of autoclaved and thoroughly washed beach sand as an inert substratum for the test organism. Survival and growth (ash free dry weight) were measured at test termination. Ash free dry weight was determined using a drying oven (105°C for at least 12 h) and a muffle furnace (550°C for at least 2 h). The ash free weight was calculated by subtracting the ashed weight (non-carbonaceous material) from the dry weight.

2.2.2. *L. stagnalis*

As no standardized test methods exist for toxicity testing with *L. stagnalis*, test methods followed those proposed by Bhargava (1992) and Slooff and Canton (1983). *L. stagnalis* were shipped by Dr. M. Grosell (RSMAS, University of Miami, Miami, FL) to the testing laboratory as egg masses and held prior to hatching in the respective site waters. Exposure treatments were prepared as described above. Nominal test concentrations ranged from 2 to 200 $\mu\text{g Ni/L}$ and varied among waters as a function of Ni bioavailability (details available in Supplemental section). Upon hatching, snails ($<24\text{ h-old}$) were

Table 1
Locations and average water quality of surface water systems where water was collected for nickel toxicity determinations. For comparison, the 10th, 50th, and 90th percentiles of pH, hardness, and DOC for European surface waters are also reported. These values were taken from ECB, 2008.

Site	Location	Hardness	Alkalinity	pH	Ca	Mg	Na	K	Cl	SO ₄	DOC ¹
		mg/L as CaCO ₃			mg/L						
Calapooia River	Albany, OR	212	200	8.0	46	21	68	1.9	73	1.2	0.69
S. Platte River	Denver, CO	256	160	7.8	71	17	120	14	120	150	7.1
S. Platte River (pH-amended)	Denver, CO	256	16	6.9	72	17	120	47	150	150	7.0
S. Santiam River	Cascadia, OR	16	24	7.1	3.4	0.81	2.4	0.25	0.98	0.51	0.94
Zollner Creek	Mt. Angel, OR	136	28	7.3	32	11	3.9	1.0	65	3.0	7.1
10th Percentile	European surface waters	16.4		6.6							2.6
50th Percentile	European surface waters	99		7.5							6.4
90th Percentile	European surface waters	253		8.1							12.4

¹ DOC = dissolved organic carbon.

randomly transferred to exposure chambers to initiate testing. Survival and growth (wet weight) were measured at test termination. Wet weight was determined to the nearest 0.01 mg (after blot-drying) using an analytical balance.

2.2.3. *B. calyciflorus*

Test methods were based on APHA standard method 8420 (APHA, 1998) and methods proposed by Snell and Moffat (1992). Rotifer (*B. calyciflorus*) test kits (ROTOXKIT F) were obtained from MicroBioTests Inc. (Ghent, Belgium). Cysts were hatched according to manufacturer's directions except that the hatching media (moderately-hard reconstituted laboratory water) and food (*Selenastrum capricornutum*) were prepared at the testing laboratory. Exposure treatments were prepared as described above. Nominal test concentrations ranged from 7.8 to 8000 µg Ni/L and varied among waters as a function of Ni bioavailability (details available in Supplemental section). Population growth rate was determined at test termination.

2.2.4. *L. minor*

Nickel toxicity for *L. minor* was assessed for plants exposed in the five natural waters, and also to APHA growth media (Environment Canada, 1999), modified to include 1 mg/L Suwannee River DOC. Further to the adjustments made to waters from Zollner Creek and S. Platte River (described above), each of the natural waters were subjected to vacuum filtration using a 47 mm, 1 µm fiber filter (Whatman GF/C), to minimize the risk of algal contamination, and the water from Santiam River was spiked with 1/6th strength APHA media (Environment Canada, 1999). This latter modification was required to achieve sufficient plant growth in control treatments as determined from preliminary investigations (data not shown).

The toxicity study with *L. minor* was based on OECD standard method 221 (OECD, 2002). Exposure treatments were prepared as described above with the exception that Ni(NO₃)₂ · 6H₂O (Acros Organics, Geel, Belgium) was used instead of NiCl₂. Nominal test concentrations ranged from 18 to 11,738 µg Ni/L and varied among waters as a function of Ni bioavailability (details available in Supplemental section). Testing was initiated by adding two *L. minor* plants, having three fronds each, to a 250 ml Erlenmeyer flask containing 100 ml of the test solution. A foam stopper was inserted into the flask and secured in place with Parafilm. To minimize plant-induced changes in the water chemistry parameters, 100% test solution renewal was performed daily. Measurements of pH were taken for both the expired test solutions and the replacement solutions and the pH of the replacement water from S. Platte was re-adjusted as necessary to pH 7.0. At test termination, the number of fronds, total root length and plant dry weight were measured. Root length was chosen to replace the OECD-recommended endpoint of total frond number because of the highly sensitive response of this endpoint to Ni²⁺ toxicity (P.M.C. Antunes, unpublished data), and to be consistent with previous Ni BLMS

for plants, which also use root growth as the measured endpoint (Lock et al. 2007; Antunes and Kreager, 2009).

Immediately after harvest, plants were placed in a drying oven for 24 h at 60 ± 2 °C. For statistical analysis, data for all endpoints were converted to specific growth rates following OECD 221 recommendations.

2.3. Chemical analysis

Water quality variables measured in the toxicity tests included dissolved oxygen (DO), temperature, pH, hardness, alkalinity, total residual chlorine, ammonia, conductivity, dissolved organic carbon (DOC), total and dissolved Ni, Ca, Mg, sodium (Na), potassium (K), chloride (Cl⁻), and sulfate (SO₄²⁻). All variables were measured at test initiation. Measurements of DO, pH, and temperature were made daily. Samples for dissolved Ni were also taken from freshly prepared test solutions prior to each renewal (*C. tentans*) or weekly prior to a renewal (*L. stagnalis*), and at test termination or complete mortality, whichever occurred first. Because of the small volume of solution used for the rotifers, no Ni samples were collected at test termination.

Measurements of total acid-extractable Ni (referred to as total Ni) and soluble Ni (sample passed through 0.45 µm nylon mesh filter following a 5-ml rinse with water sample; referred to as dissolved Ni) were made in the control water and each treatment. Samples were collected into polypropylene conical tubes and acidified to pH < 2 with concentrated nitric acid (ACS grade; Mallinckrodt, Phillipsburg, NJ, USA) before analysis by graphite or flame atomic absorption spectroscopy (AA800; Perkin-Elmer, Shelton, CT, USA). All analyses included appropriate quality assurance samples (e.g., duplicates, matrix spikes, reference standards).

Samples collected for DOC analysis were passed through a 0.45 µm nylon mesh filter following a 5-ml rinse with water sample, acidified with concentrated H₂SO₄, and stored in amber glass bottles. DOC concentrations were determined via combustion using US EPA method 415.2. Other analytes were analyzed by inductively coupled-plasma atomic emission spectroscopy (ICP-AES) (Ca, Mg, Na, and K; US EPA method 200.7), ion chromatography (SO₄²⁻; US EPA method 300.1) or colorimetric titration (Cl⁻, hardness and alkalinity). Water pH and ammonia were measured using ion-selective electrodes (Thermo Electron, Beverly, MA, USA). Conductivity and total residual chlorine were measured with portable probes from Hach Co. (Loveland, CO, USA). Dissolved oxygen was measured using a DO probe which was calibrated in air-saturated water (Yellow Springs Instruments, Yellow Springs, OH, USA).

2.4. Toxicity test data analysis

Chronic effect concentrations that represented 10, 20 or 50% reduced performance relative to controls (EC10, EC20 or EC50, respectively) and 95% confidence limits were calculated using TRAP (Toxicity Relationship Analyses Program, Version 1.10., US EPA, NHERL, Duluth, MN USA). The

toxicant concentrations used in these calculations were averages of all dissolved measurements for each treatment within each toxicity test.

2.5. BLM modeling

The accuracy of the model predictions was evaluated by comparing the observed toxicity for each of the non-BLM species with the predicted toxicity from the BLMs. Predictions for the invertebrate species were made with the chronic Ni BLMs for *C. dubia* (De Schamphelaere et al., 2006a) and *D. magna* (Deleebeek et al., 2008), For *L. minor*, BLM predictions were performed using the chronic Ni BLMs for *P. subcapitata* (Deleebeek et al., 2009a) and *H. vulgare* (Lock et al., 2007). Equations and parameter values for these models are shown in Table 2. All BLMs used the Windermere Humic Aqueous Model VI (WHAM-Model VI, Tipping, 1998), to calculate metal speciation, with stability constants for inorganic metal complexes taken from the NIST database. Following recommendations of Van Laer et al. (2006), who calibrated WHAM-Model VI with measured Ni speciation data in natural waters, we set the input of the fulvic acid (FA) concentration (mg/L) for the modeling at 0.8 times the measured DOC concentration, we used a Ni-FA binding constant $\log K_{MA} = 1.75$, and we assumed that Fe^{3+} and Al^{3+} activities were controlled by their colloidal $Fe(OH)_3$ and $Al(OH)_3$ precipitates. All these assumptions were consistent with the assumptions made for the development of the BLMs in the original papers as cited above. Predictions were made of EC50 or EC20 values. The full water chemistry used for BLM predictions is provided in the Supplemental section as supportive information.

2.5.1. Determining accuracy of BLMs

Under the assumption that the BL-parameters, which were obtained for BLM organisms and which describe competitive binding of Ca^{2+} , Mg^{2+} , H^+ and Ni^{2+} for the biotic ligand, are constant across organisms, we fitted the intrinsic sensitivity (see Table 2) of each organism that resulted in the lowest sum of log-deviations between observed and predicted ECx values ($\log\text{-deviation} = \text{abs}[\log(\text{ob-}$

served ECx) – $\log(\text{predicted ECx})$ using the available toxicity data for all waters. The magnitude of prediction errors is then a measure of how well the existing BLM parameters obtained from “BLM organisms” can explain the among-test-water variability in dissolved ECx values of non-BLM organisms that can be attributed to differences in chemistry. This procedure is identical to the one used in original BLM studies (Di Toro et al., 2001, Santore et al., 2001).

2.5.2. Analysis of reduction of intra-species variability

The ability of the BLMs to reduce intra-species variability was assessed by comparing intra-species variability between non-normalized toxicity data and data that were normalized to a common set of abiotic factors. Since this aim is slightly different than in the previous section, the modeling procedure is slightly different too (although built on exactly the same principle). In the previous exercise, a single global intrinsic sensitivity was fitted per species to toxicity data in all waters; here, each test water is considered to yield a separate intrinsic sensitivity. Thus, the variation of intrinsic sensitivities within a species but among waters reflects residual variation (e.g. due to temporal variability of sensitivity of lab test populations) and variation which is due to the fact that the BLM parameters describing competitive binding are not 100% the same across organisms. When each of these ‘water-specific’ intrinsic sensitivities is then translated to a common set of abiotic factors (using the equations in Table 2), this remaining variation can be expressed as a variation of dissolved ECx values. The entire procedure is called normalization and has also been used in risk assessments of Cu and Zn (Van Sprang et al., 2009). Thus, the reduction of the variability from non-normalized to BLM-normalized ECx values can be quantified. Intra-species variability was expressed as the ratio between the highest and lowest non-normalized or normalized EC20 or EC50 value for a specific species, i.e., the max/min ratio. Each ECx value for each organism was normalized individually to the mean pH (7.8), hardness (217 mg $CaCO_3/L$), and DOC (2.8 mg/L) of the Rhine River, which represents a typical European surface water system. The full chemistry that was used for modeling is included as supportive information.

3. Results

3.1. Water chemistry

Water chemistry data for the sites evaluated in the present study are reported in Table 1. The S. Santiam River (hardness of 16 mg/L as $CaCO_3$, pH of 7.1, DOC of 0.9 mg/L) was representative of a system with high Ni bioavailability and was expected to result in the greatest nickel toxicity. The range of pH and hardness among these waters is very close to the 10th and 90th percentile of the distributions of these parameters in European surface waters (Table 1). DOC concentration from the S. Santiam River (0.9 mg/L) was below the 10th percentile of DOC reported for European surface waters (2.6 mg/L), and the DOC concentration for the South Platte River (7.1 mg/L) was below the 90th percentile of DOC reported for European surface waters (12.4 mg/L). The pH-adjusted South Platte River (hardness of 256 mg/L as $CaCO_3$, pH of 6.9, DOC of 7.1 mg/L) was representative of a system with low Ni bioavailability and was expected to result in the lowest nickel toxicity. The remaining waters were expected to be in the mid-range in terms of nickel toxicity.

3.2. Toxicity testing

C. tentans was relatively insensitive to chronic Ni exposure with 10-day EC20 values ranging from 372 μg Ni/L for S. Santiam River water to 1029 μg Ni/L for the S. Platte River (Table 3). In general, Zollner Creek and the S. Santiam River yielded the lowest EC10 and EC20 values, whereas the S. Platte River yielded the highest values.

Table 2

Equations and parameter values of chronic Ni Biotic Ligand Models for the BLM-species *Ceriodaphnia dubia* (De Schamphelaere et al., 2006a) *Daphnia magna* (Deleebeek et al., 2008), *Pseudokirchneriella subcapitata* (Deleebeek et al., 2009a) and *Hordeum vulgare* (Lock et al., 2007) (including intrinsic sensitivities for these species), and fitted intrinsic sensitivities for non-BLM species (this study).

Model parameters BLM-species	Invertebrate models ¹		Plant/algae models ²	
	<i>Daphnia magna</i>	<i>Ceriodaphnia dubia</i>	<i>Pseudokirchneriella subcapitata</i>	<i>Hordeum vulgare</i>
$\log K_{CaBL}$	3.53	3.53	2.0	–
$\log K_{MgBL}$	3.57	3.57	3.3	3.47
$\log K_{HBL}$	–	–	6.5	–
S_{pH}	0.3335	0.8587	–	–
Intrinsic sensitivity (using invertebrate models)				
$Q_{50, Daphnia}^*$	3.986	–	–	–
$Q_{20, Daphnia}^*$	4.094	–	–	–
$Q_{50, Ceriodaphnia}^*$	–	1.321	–	–
$Q_{20, Ceriodaphnia}^*$	–	1.447	–	–
$Q_{20, Chironomus}^*$	3.353	–0.612	–	–
$Q_{20, Brachionus}^*$	3.265	–0.794	–	–
$Q_{50, Lymnaea}^*$	5.091	1.099	–	–
$Q_{50, Lemna}^*$	3.743	–0.436	–	–
Intrinsic sensitivity (using algae/plant models)				
$EC50_{Ni^{2+}}^*$	–	–	1.23 $\mu mol/L$	–
$EC50_{Ni^{2+}}^*$ <i>Pseudokirchneriella</i>	–	–	–	7.12 $\mu mol/L$
$EC50_{Ni^{2+}}^*$ <i>Hordeum</i>	–	–	0.876 $\mu mol/L$	0.778 $\mu mol/L$

¹ Predicted $ECX_{Ni^{2+}} = 10^{-(Q_x^* + S_{pH} \times pH) \times \{1 + K_{CaBL} \cdot (Ca^{2+}) + K_{MgBL} \cdot (Mg^{2+})\}}$.

² Predicted $ECX_{Ni^{2+}} = ECX_{Ni^{2+}}^* \cdot \{1 + K_{CaBL} \cdot (Ca^{2+}) + K_{MgBL} \cdot (Mg^{2+}) + K_{HBL} \cdot (H^+)\}$.

Table 3
Chronic effect concentrations ($\mu\text{g Ni/L}$) for *Chironomus tentans* (ash free dry weight), *Lymnaea stagnalis* (wet weight), *Brachionus calyciflorus* (population growth rate), and *Lemna minor* (average specific growth rate based on root length). 5th and 95th percent confidence intervals are presented in parentheses. Effect concentrations on the basis of free ionic Ni^{2+} activity are given as supportive information.

Source water	<i>Chironomus tentans</i>		<i>Lymnaea stagnalis</i>		<i>Brachionus calyciflorus</i>		<i>Lemna minor</i>	
	EC ₁₀	EC ₂₀	EC ₁₀	EC ₂₀	EC ₁₀	EC ₂₀	EC ₁₀	EC ₅₀
Calapooia River	404.3 (NA ¹ –550.3)	544.3 (NA–748.1)	1.1 (0.9–1.4)	1.6 (1.4–2.3)	662.6 (282.8–1018.4)	990.4 (627.9–1457.8)	7.5 (2.8–20.1)	228.8 (138.8–377.3)
S. Platte River	782.2 (381.4–1241.2)	1028.8 (700.3–1998.6)	19.5 (4.1–27.5)	26.8 (12.4–31.3)	945.2 (55.3–1213.4)	1141.1 (502.6–1626.7)	73.9 (35.5–153.7)	550.9 (403.8–751.5)
S. Platte River (pH-amended)	405.4 (302.3–3365.1)	551.0 (354.5–4831.2)	7.0 (4.8–55.4)	21.3 (6.9–62.2)	4.7 (2.9–11.2)	7.4 (3.8–10.2)	435.3 (287.4–659.3)	1377 (1162.6–1630.9)
S. Santiam River	305.7 (211.4–378.7)	372.3 (306.6–4752)	1.4 (0.8–12.2)	6.3 (1.3–14.1)	103.9 (91.3–121.0)	143.9 (125.2–178.4)	8.2 (4.8–14.2)	87 (68.7–110)
Zollner Creek	251.1 (164.7–484.9)	501.7 (328.9–1690.0)	1.3 (1.1–1.8)	1.9 (1.7–3.0)	1379.3 (1207.3–1390.8)	1497.6 (1345.5–1522.3)	36.0 (22.1–351.5)	284.4 (230.1–351.5)
APHA Media	NC ²	NC ²	NC ²	NC ²	NC ²	NC ²	3.9 (1.7–8.7)	136.8 (94.8–197.5)

¹ NA: not available. The 5th percent confidence interval was not reliable.

² NC: not calculated. APHA media was used only for *Lemna minor* toxicity tests.

L. stagnalis was the most sensitive to Ni of the invertebrates tested in the natural waters, with EC10 values $<2 \mu\text{g/L}$ dissolved Ni (based on wet weight) for Calapooia River, Zollner Creek and S. Santiam River (Table 3). Nickel toxicity varied considerably among the sites, with EC20 values ranging from $1.6 \mu\text{g Ni/L}$ for Calapooia River to $26.8 \mu\text{g Ni/L}$ for the ambient S. Platte River.

Although *B. calyciflorus* was on average the least sensitive to Ni of the invertebrate organisms tested, it had the largest range in observed effects. For example, the overall range of EC10 values was 4.7 to $1379 \mu\text{g/L}$ dissolved Ni based on population growth rate (Table 3). The pH-amended S. Platte River showed the highest Ni toxicity, with EC10 and EC20 values of $<10 \mu\text{g/L}$ dissolved Ni. Zollner Creek, on the other hand, was consistently the most protective water with EC10 and EC20 values approaching nearly $1500 \mu\text{g/L}$ dissolved Ni (Table 3).

Lowering the pH in the S. Platte River dramatically increased Ni toxicity to *B. calyciflorus*, which was unexpected and not consistent with the observed results for *L. stagnalis* and *C. tentans*. An effort was therefore made to explain the high sensitivity of *B. calyciflorus* to the pH-amended S. Platte River water. First, the results obtained for the pH-amended water were confirmed in a second test (data not shown) and this verified that the low value was not anomalous. Second, three toxicity tests were conducted with reconstituted laboratory water to determine whether the observed differences were due to exposure to CO_2 (which was used to control the pH in the original experiment) or to the presence of pH-sensitive contaminants in the natural water. The toxicity tests were performed by making reconstituted laboratory waters that mimicked the S. Platte River in hardness (including Ca to Mg ratio), alkalinity, pH and DOC (isolated natural organic matter from the Black River, SC, USA, using reverse osmosis (Ryan et al, 2009)). The reconstituted water was tested with and without pH-amendment (similar to site water tests). The third test was performed at an extremely low alkalinity (8 mg/L as CaCO_3 ; achieved using NaHCO_3) in order to achieve a low pH level without the need for a CO_2 enriched atmosphere.

The EC10 for the pH-amended reconstituted S. Platte water was $1009 \mu\text{g Ni/L}$ (Table 4), which was comparable to the EC10 of $946.4 \mu\text{g Ni/L}$ for the ambient pH reconstituted S. Platte water (Table 4) and the EC10 of $945.2 \mu\text{g Ni/L}$ for the test performed on natural S. Platte River water at ambient pH (Table 4). The EC10 from the low alkalinity reconstituted water (without CO_2 enrichment) was $498.3 \mu\text{g Ni/L}$, which is lower than in the pH-amended reconstituted water (with CO_2 enrichment). This suggests that exposure to CO_2 was not the principle source of toxicity for the pH-amended natural S. Platte River water. These results suggest that the high toxicity of Ni to *B. calyciflorus* in the pH-adjusted S. Platte River was caused by pH-sensitive contaminants in the river, as opposed to increased Ni toxicity at the lower pH. Data from the pH-adjusted S. Platte River treatment were therefore not used further in the BLM modeling for any species.

For *L. minor*, average growth rate inhibition (based on root length) varied considerably among the test waters. For example, EC10s and EC50s varied 111 and 15 fold, respectively (Table 3). The lowest EC10 was observed for the modified APHA medium, whereas the lowest EC50 was observed for the S. Santiam River. The least sensitive water was the pH-adjusted S. Platte River water.

3.3. Accuracy of BLM predictions

Predictions for *C. tentans* were equally accurate with either the *D. magna* or the *C. dubia* BLMs, both of which predicted EC20s for all waters with a less than 2-fold error and with an average error of 1.6-fold (Fig. 1A,B).

Predictions for *L. stagnalis* were better with the *C. dubia* BLM than with the *D. magna* BLM (Fig. 2A,B). The EC50s for all waters were predicted within a factor of 1.8 using the *C. dubia* BLM with an average prediction error of 1.5-fold. With the *D. magna* BLM, prediction errors were as high as 2.4 fold, with an average of 1.8-fold.

Table 4

Chronic toxicity of Ni to *Brachionus calyciflorus* in natural and reconstituted waters. Toxicological data for site water tests taken from Table 3. Dissolved organic carbon (DOC) and point estimate concentrations for reconstituted waters are based on nominal concentrations. Effect concentrations on the basis of free ionic Ni^{2+} activity are given as supportive information.

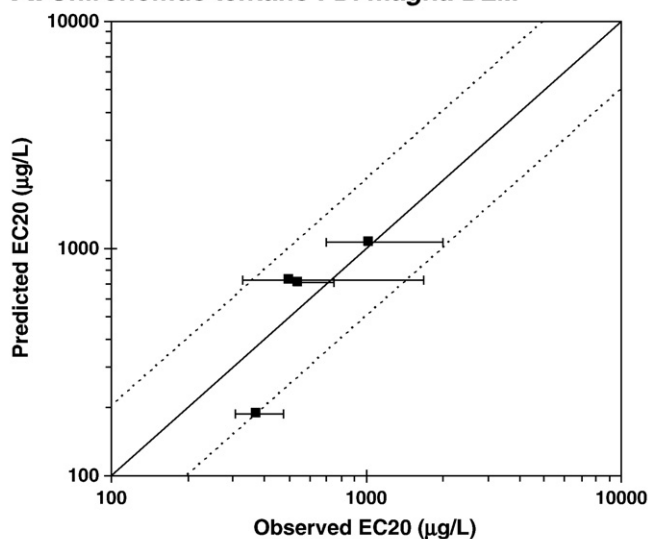
Water	Hardness	Alkalinity	pH	DOC	EC10	EC20
	mg/L as CaCO_3				$\mu\text{g/L}$ dissolved Ni	
S. Platte natural – ambient pH	256	160	7.8	7.1	945.2 (55.3–1213.4)	1141.1 (502.6–1626.7)
S. Platte – pH-amended	256	16	6.9	7.0	4.7 (2.9–11.2)	7.4 (3.8–10.2)
S. Platte reconstituted – ambient pH	270	190	8.2	7.0	946.4 (592.9–1520.4)	1422.7 (945.1–2141.0)
S. Platte reconstituted – pH-amended	260	14	7.1	7.0	1009.0 (257.5–3954.4)	1432.0 (439.9–4656.0)
S. Platte reconstituted – low alkalinity	280	8	7.1	7.0	498.3 (108.4–2288.9)	787.7 (234.3–2648.0)

Predictions for *B. calyciflorus* were clearly better with the *D. magna* BLM (Fig. 3A) than with the *C. dubia* BLM (Fig. 3B). Predicted EC20s for all waters were accurate within a factor of 2.2 using the *D. magna* BLM with an average prediction error of 1.8-fold. With the *C. dubia* BLM,

the predictions differed by up to 4.6-fold from observation, with an average error of 2.9-fold.

Both the *P. subcapitata* BLM (Fig. 4A) and the *H. vulgare* BLM (Fig. 4B) predicted all the EC50s of *L. minor* within a factor of 2.4, with an average error of 1.6-fold. Since the *P. subcapitata* model uses H^+ , Mg^{2+} , and DOC to explain variability in toxicity, and the *H. vulgare* model uses only Mg^{2+} and DOC, we wished to determine if the

A. *Chironomus tentans* : *D. magna* BLM



B. *Chironomus tentans* : *C. dubia* BLM

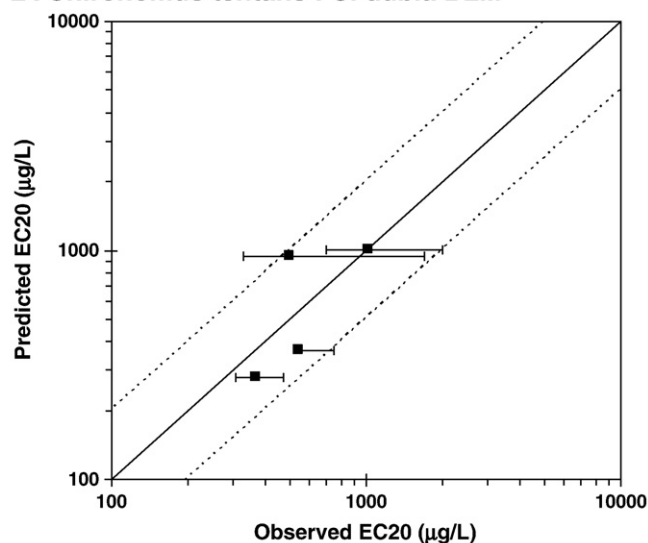
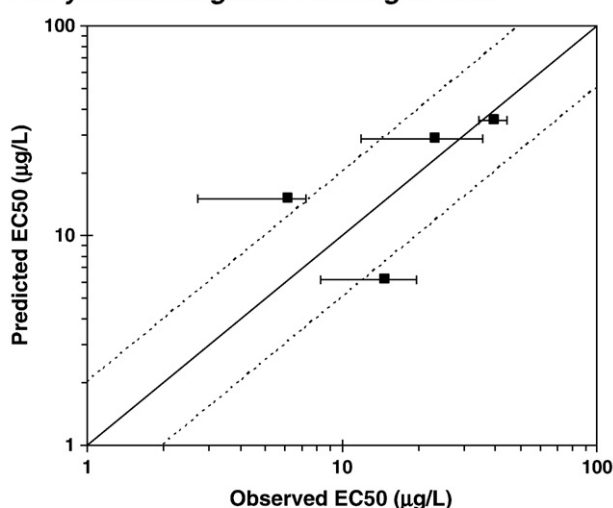


Fig. 1. Observed nickel toxicity (EC20, in $\mu\text{g Ni/L}$) to the insect *Chironomus tentans* compared with predicted toxicity using the Biotic Ligand Model developed for *Daphnia magna* (A) and *Ceriodaphnia dubia* (B). The error bars represent 95% confidence intervals. The solid line (—) represents a 1:1 relationship. The dashed line (---) represents predictions within a factor of 2.

A. *Lymnaea stagnalis* : *D. magna* BLM



B. *Lymnaea stagnalis* : *C. dubia* BLM

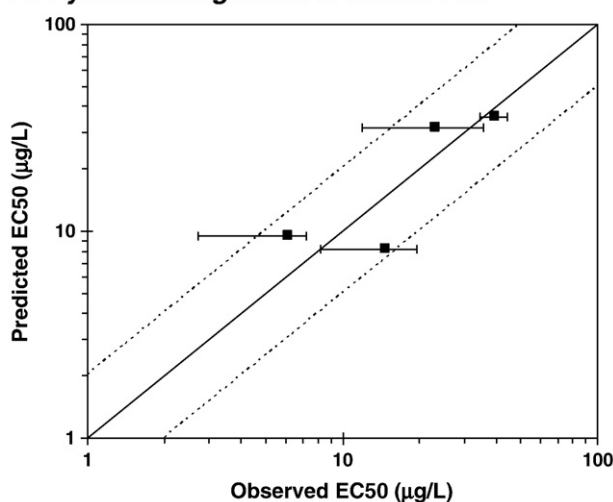


Fig. 2. Observed nickel toxicity (EC50, in $\mu\text{g Ni/L}$) to the snail *Lymnaea stagnalis* compared with predicted toxicity using the Biotic Ligand Model developed for *Daphnia magna* (A) and *Ceriodaphnia dubia* (B). The error bars represent 95% confidence intervals. The solid line (—) represents a 1:1 relationship. The dashed line (---) represents predictions within a factor of 2.

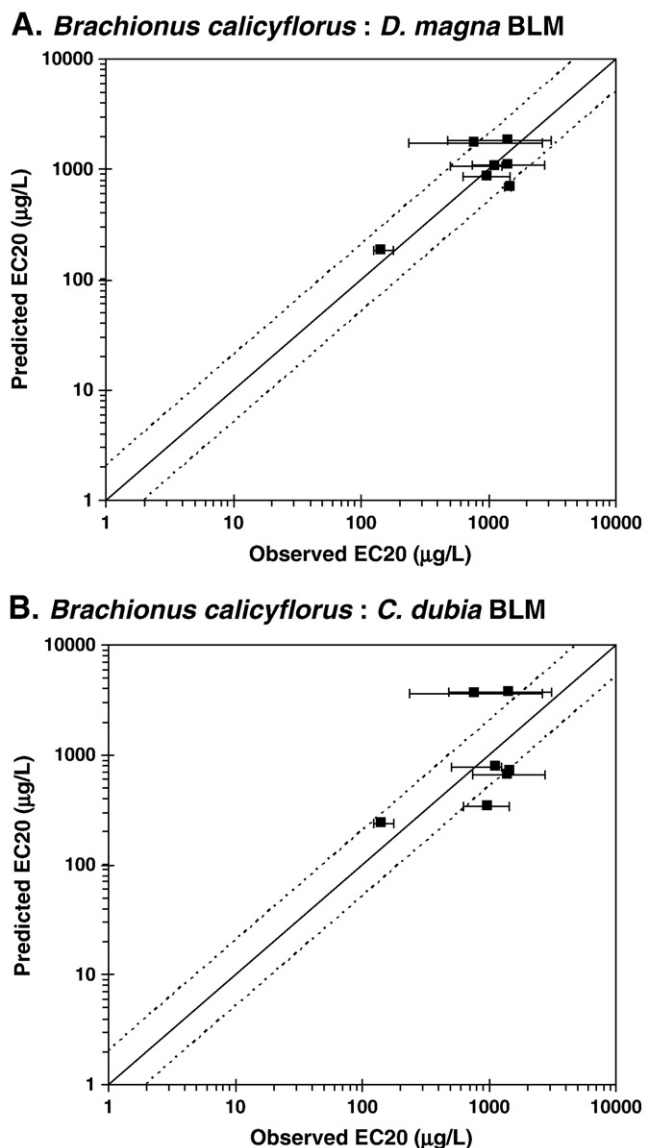


Fig. 3. Observed nickel toxicity (EC20, in $\mu\text{g Ni/L}$) to the rotifer *Brachionus calyciflorus* compared with predicted toxicity using the Biotic Ligand Model developed for *Daphnia magna* (A) and *Ceriodaphnia dubia* (B). The error bars represent 95% confidence intervals. The solid line (—) represents a 1:1 relationship. The dashed line (---) represents predictions within a factor of 2.

addition of a Ca^{2+} interaction would improve the model accuracy. Therefore predictions using the BLM parameters developed for *D. magna* (Fig. 4C) and *C. dubia* (Fig. 4D) were compared to the outcome of the *L. minor* toxicity tests. Both of these models incorporate Ca^{2+} , Mg^{2+} and H^+ in their calculations of predicted nickel toxicity. Predictions from the *D. magna* and *C. dubia* BLMs were within a factor of 1.6 and 1.8, respectively, with average prediction errors of 1.3 and 1.4, respectively. Generally, these predictions are superior to those obtained with the plant and algae BLMs and could suggest a significant Ca^{2+} interaction with Ni toxicity in *L. minor*.

3.4. Reduction of intra-species variability

The chronic Ni BLMs substantially reduced the observed intra-species variability for three of the four non-BLM species (Fig. 5). The exception was *C. tentans*, for which the max/min ratio of normalized data was similar (3.5% higher) than the non-normalized data. However, the intra-species variability for *C. tentans* was low to begin with (i.e.,

similar to the variability for *L. stagnalis* and *L. minor* after normalization) and remained low after normalization. Intra-species variability was substantially reduced for the two species with the most variable ECx values, *B. calyciflorus* and *L. stagnalis*. Intra-species variability was reduced by a factor of 3.4 for *B. calyciflorus* using the *D. magna* BLM and by a factor of 2.3 for *L. stagnalis* using the *C. dubia* BLM. The use of the *P. subcapitata* BLM slightly reduced the intra-species variability for *L. minor*; however, variability was reduced by a factor of 2.4 when the *C. dubia* or *D. magna* BLM were used for *L. minor*.

4. Discussion

An important question in the use of toxicity-based bioavailability models for metals within chemicals management frameworks is whether or not models developed for one species can be applied to other species within ecotoxicity databases (Van Sprang et al., 2009). Confirmation of the cross-species extrapolation approach would streamline the process by which ecotoxicity data for numerous species can be normalized to site-specific abiotic conditions, which provides enormous benefits in terms of site-specific risk assessment and the derivation of flexible environmental quality standards. In the present study, confirming that BLMs developed for one species can be used to normalize ecotoxicity data for other species suggest that intrinsic sensitivity is species-specific, but interactions between relevant abiotic factors and the free nickel ion at the biotic ligand are reasonably similar across species. Previous studies provide evidence to support cross-species extrapolation of chronic Ni BLMs for fish (Deleebeeck et al. 2007a), algae (Deleebeeck et al., 2009b, Worms and Wilkinson, 2007), and cladocera (Deleebeeck et al. 2007b), as well as for chronic Cu (De Schampelaere et al., 2006b) and Zn BLMs (Van Sprang et al., 2009). The results from the present study provide further evidence to support the extrapolation of nickel BLMs to two important groups, non-cladoceran invertebrates and vascular plants.

Two criteria were used in the current study to evaluate the validity of extrapolating chronic nickel BLMs to non-BLM invertebrates (*C. tentans*, *L. stagnalis* and *B. calyciflorus*) and a vascular plant (*L. minor*). The first criterion was the accuracy with which BLMs predicted Ni toxicity to the non-BLM organisms. The *D. magna* and *C. dubia* BLMs showed average prediction errors of less than 1.6-fold. The ranges of pH and hardness in waters used for toxicity testing fell within the 10th and 90th percentiles of the distributions of these parameters in European surface waters, whereas DOC ranged from below the 10th percentile to above the 50th percentile of European surface waters (Table 1). While additional geochemical combinations would have provided useful information, it was not feasible to include additional waters due to the effort involved. Considering the substantial differences in geochemistry among waters tested, and that toxicity varied up to a factor of 13 (e.g., EC50 values for *L. stagnalis*, Table 2), the results (Figs. 1–3) suggest that competitive binding of Ca^{2+} , Mg^{2+} , H^+ and Ni^{2+} for the biotic ligand are sufficiently similar between BLM organisms (*D. magna* and *C. dubia*) and non-BLM organisms (*C. tentans*, *L. stagnalis* and *B. calyciflorus*) to support cross-species extrapolation of the chronic invertebrate Ni BLMs. However, univariate experiments are necessary to specifically define the individual effects of Ca, Mg, and pH on chronic Ni toxicity, since the current dataset with natural waters does not allow the relative importance of competition effects vs. speciation effects (notably, complexation with DOC) to be directly determined.

For *L. minor*, the *D. magna* and *C. dubia* BLMs showed the most accurate predictions, which suggest a significant Ca^{2+} interaction with Ni toxicity for *L. minor* (Fig. 4). These results are supported by recently published work of Antunes and Kreager (2009) who generated toxicity predictions for *H. vulgare* within a factor of (+/–) 1.5 using a BLM that includes both H^+ and Ca^{2+} interactions. Their work shows that readjustment of the relative magnitude of the BLM binding parameters (and not just the addition of a Ca-binding term to the BLM for *H. vulgare*

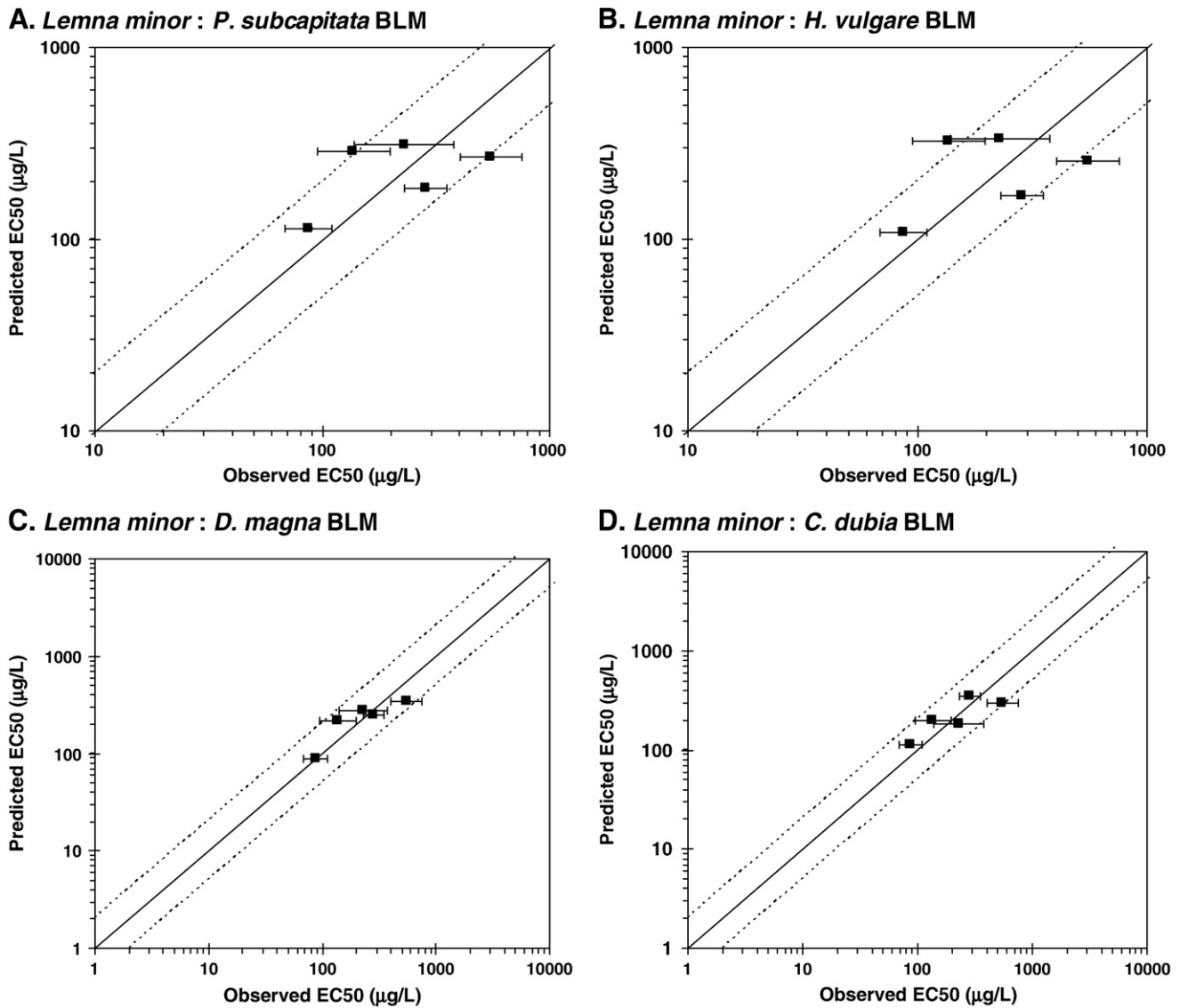


Fig. 4. Observed nickel toxicity (EC50, in $\mu\text{g Ni/L}$) to the vascular plant *Lemna minor* compared with predicted toxicity using the Biotic Ligand Model developed for the alga *Pseudokirchneriella subcapitata* (A), the vascular plant *Hordeum vulgare* (B), and the cladocera *Daphnia magna* (C) and *Ceriodaphnia dubia* (D). The error bars represent 95% confidence intervals. The solid line (—) represents a 1:1 relationship. The dashed line (---) represents predictions within a factor of 2.

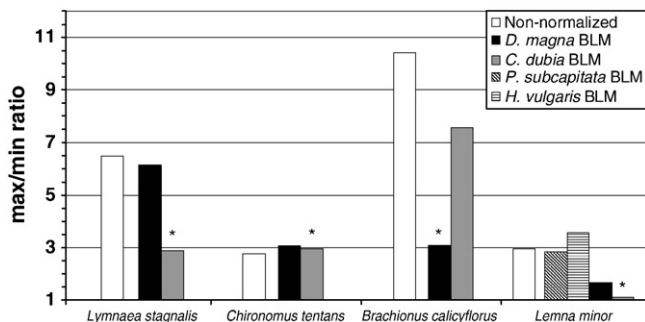


Fig. 5. Intra-species variability (expressed as max/min ratios) of nickel EC20 (*Brachionus calyciflorus* and *Chironomus tentans*) or EC50 concentrations (*Lymnaea stagnalis* and *Lemna minor*) (dissolved $\mu\text{g Ni/L}$) expressed as non-normalized values and Biotic Ligand Model (BLM)-normalized values. The data from all species were normalized using BLMs for the cladocera *Daphnia magna* and *Ceriodaphnia dubia*. *Lemna minor* toxicity data were also normalized using BLMs for the alga *Pseudokirchneriella subcapitata* and the vascular plant *Hordeum vulgare*. The asterisks indicate the BLM chosen for the derivation of Predicted No Effects Concentrations for nickel.

of Lock et al. 2007 or the Algal BLM of Worms and Wilkinson, 2007) may improve the overall predictive accuracy of the model – a finding consistent with the better fit provided by the *D. magna* and *C. dubia* BLMs.

The second criterion was the extent to which the BLMs reduced intra-species variability in non-BLM organisms when effect concentrations in all media were normalized to an average EU condition. This analysis showed that the BLMs are indeed capable of reducing the influence of abiotic parameters (Fig. 5). In this study, substantive reductions in intra-species variability were observed for both sensitive species, e.g., *L. stagnalis* (2.3-fold reduction) and *L. minor* (2.6-fold reduction), and insensitive species, e.g., *B. calyciflorus* (3.4-fold reduction).

4.1. Species sensitivity comparison

While we do acknowledge that slopes of concentration response curves may vary between different species and that, as a consequence, species sensitivity ranks may be different depending on the effect level chosen for performing the comparison, our discussion here is limited to a sensitivity comparison based on EC10 values due to the relevance of this effect level to EU regulation. A wide range of sensitivities was

observed among the species tested in the present study (EC10 values ranged from 1.1 µg Ni/L for *L. stagnalis* to 1379.3 µg Ni/L for *B. calyciflorus*). The greater range in Ni sensitivity among invertebrate species tested in the present study (EC10 values of between 1.1 µg Ni/L for *L. stagnalis* and 1379.3 µg Ni/L for *B. calyciflorus*), compared to the range reported for species in the chronic Ni ecotoxicity database (EC10 values of between 2.8 µg Ni/L for *C. dubia* to 256 µg Ni/L for *D. magna* (ECB, 2008)), supports the relevance of the organisms selected. Few data are available for the species tested in the present study, and those that are available are consistent with the results we obtained. For example, the results of tests with *C. tentans* support the previous results indicating that *Chironomus* sp. are not sensitive to Ni exposure. The 10-day exposures with *C. tentans* in the current study agree with a 30-day growth experiment with *Chironomus riparius* conducted at a hardness and pH of 260 mg/L as CaCO₃ and 8.6, respectively (estimated NOEC = 1100 µg/L dissolved Ni) (Powlesland and George, 1986). A notable finding of the present study is that *L. stagnalis* is considerably more sensitive to Ni than other snails. For example, *Juga plicifera* had a reported 21-day no-observable effects concentration (NOEC) for survival of 124 µg/L dissolved Ni in a relatively soft well water (hardness and pH of 59 mg/L as CaCO₃ and 7.1, respectively) (Nebeker et al., 1986). This NOEC is well above the highest EC10 reported for the growth endpoint of *L. stagnalis* (19.5 µg/L, Table 3).

The EC10 values of the four non-BLM species were normalized to pH of 7.8, hardness of 217 mg CaCO₃/L, and DOC of 2.8 mg/L, and these normalized EC10 values were compared with those already included in the Nickel ecotoxicity database (ECB, 2008). A comparison of the new and accepted species shows that *B. calyciflorus* and *C. tentans* would be among the least sensitive organisms for these water quality conditions (Table 5), with sensitivity rank positions of 29 and 30. In terms of sensitivity, *B. calyciflorus* and *C. tentans* rank just below *Bufo terrestris*, which is the least sensitive of all species accepted in the database. Therefore, the fact that the BLMs did not affect the intra-species variability for *C. tentans* has little bearing on the outcome of the SSD for this scenario.

More importantly, when the EC10 data from this study are included in the Nickel ecotoxicity database (ECB, 2008), the most sensitive species in this scenario will be *L. stagnalis*, which will replace *C. dubia* (Table 5). The *C. dubia* BLM was able to substantially decrease the intra-species variability for *L. stagnalis*, so the influence of abiotic factors on the outcome of the SSD will be taken into account when the effects data for *L. stagnalis* are normalized to site-specific water chemistries.

Table 5

Sensitivity ranking of the newly tested non-BLM species (*Brachionus calyciflorus*, *Chironomus tentans*, *Lymnaea stagnalis*, and *Lemna minor*) compared with a selection of the 31 species reported in Nickel Aquatic Effects Assessment (ECB, 2008), including *Ceriodaphnia dubia*, *Lemna gibba*, and *Bufo terrestris*. EC10^a values (µg Ni/L) were normalized to the pH = 7.8, hardness = 217 mg CaCO₃/L, DOC = 2.8 mg/L. Species are ranked in terms of sensitivity, with 1 being the most sensitive, and 31 being the least sensitive.

Species	BLM used	Normalized EC10 (µg Ni/L, geometric mean) ^b	Sensitivity rank
<i>Ceriodaphnia dubia</i>	<i>C. dubia</i>	8.5	2
<i>Brachionus calyciflorus</i>	<i>D. magna</i>	840	30
<i>Chironomus tentans</i>	<i>C. dubia</i>	415	29
<i>Lymnaea stagnalis</i>	<i>C. dubia</i>	2	1
<i>Lemna minor</i>	<i>P. subcapitata</i>	12.4	3
<i>Lemna minor</i>	<i>D. magna</i>	20.0	3
<i>Lemna gibba</i>	<i>P. subcapitata</i>	28.5	6
<i>Bufo terrestris</i>	<i>O. mykiss</i>	1283	31

^a EC10 values have not been screened according to the approaches outlined in the Aquatic Effects Assessment of Nickel.

^b Per species the geometric mean is given of all individually (i.e. per test water) normalized EC10 values (i.e. those reported in Tables 3 and 4, with the exception of the South Platte River natural water with adjusted pH). For *Ceriodaphnia dubia*, *Lemna gibba* and *Bufo terrestris*, the values were directly taken from the effects assessment (ECB, 2008).

4.2. Implications for risk assessment and environmental quality standard derivation

Our data suggest that both the *D. magna* and *C. dubia* BLM are suitable for normalizing toxicity data for the organisms tested in the present study, although the degree of normalization by these two models was not the same for each invertebrate. Based on which model showed the least intra-species variability, we suggest that the *C. dubia* model be used for *L. stagnalis*; the *D. magna* model for *B. calyciflorus*; and either the *C. dubia* or *D. magna* models for *C. tentans* and *L. minor* (as both models provided equally good normalization for these species). For invertebrates that have not been tested here, but are in the Nickel ecotoxicity database (ECB, 2008), we suggest that the normalizations be carried out with both models and that a sensitivity analysis be carried out to determine the extent to which the choice of models affects the outcome of the SSD analysis (e.g., the hazardous concentration at the 5th percentile, or the HC5).

A full bioavailability normalization approach was used in the recently completed Existing Substances Risk Assessment of Nickel (ECB, 2008), which was a regulatory assessment of the risks associated with the ongoing production and use of nickel in the European Union (EEC 793/93). Regional Predicted No Effects Concentrations (PNECs) for Ni were calculated using ecotoxicity data for 31 aquatic species (i.e., the Nickel ecotoxicity database) that were normalized to distributions of pH, hardness, and DOC from seven typical European freshwater systems. Normalization of invertebrate ecotoxicity data followed the normalization scheme identified above, with the exception that the more stringent of the invertebrate BLMs was chosen for invertebrates in the Nickel ecotoxicity database for which no BLM exists, and that were not tested in the present study. Ecotoxicity data for algae were normalized using the *P. subcapitata* BLM, and fish and amphibian data were normalized using the *O. mykiss* BLM. The most sensitive system, with a PNEC of 3.6 µg Ni/L, had a combination of relatively high pH (7.7) and low DOC (2.5 mg/L). Much less sensitive, with a PNEC of 21.8 µg Ni/L, was a system with low pH (7.0) and high DOC (12 mg/L). This broad range of PNEC values shows the need for flexible, bioavailability-based environmental quality standards for Ni.

5. Conclusion

This study suggests that it is justified to perform a cross-species normalization of the Nickel Aquatic Effects Database by extrapolating chronic Ni BLMs to non-BLM species. This conclusion is based on the ability of chronic nickel BLMs that were developed for *C. dubia* and *Daphnia magna* to explain differences in chronic nickel toxicity that were observed in five natural waters for three invertebrates (*Brachionus calyciflorus*, *C. tentans*, and *L. stagnalis*) and one vascular plant (*L. minor*) for which no existing nickel ecotoxicity data were available. The observed nickel toxicities were variable among the test waters, especially for *L. stagnalis* and *L. minor*. Predicted chronic nickel toxicities based on the nickel BLMs were within a factor of 2 of observed toxicity in 17 out of 18 cases. Furthermore, the BLMs reduced the intra-species variability among the different waters, particularly for the most sensitive species, *L. stagnalis* and *L. minor*. Taken together, the invertebrate chronic nickel BLMs showed the ability to account for differences in toxicity to the species tested that were caused by site-specific differences in pH, hardness, and dissolved organic carbon. This study compliments the results of studies for fish (Deleebeeck et al., 2007a,b) and algae (Worms and Wilkinson, 2007), and together support the cross-species extrapolation of chronic nickel BLMs to non-BLM species in the Nickel Aquatic Effects Database when making site-specific determinations about nickel toxicity to freshwater organisms.

Supplementary materials related to this article can be found online at doi:10.1016/j.scitotenv.2010.09.012.

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References

- American Public Health Association. Standard methods for examination of water and wastewater. In: Clesceri L, Greenburg AE, Eaton AD, editors. 20th Ed. Method 8420 C. Baltimore, MD: United Book Press Inc.; 1998.
- Antunes PMC, Kreager NJ. Development of the terrestrial biotic ligand model for predicting nickel toxicity to barley (*Hordeum vulgare*): ion effects at low pH. *Environ Toxicol Chem* 2009;28:1704–10.
- Bhargava S. General behavior and mortality of *Lymnaea stagnalis* and their egg masses under the stress of thiourea and DDT. *J Fresh Biol* 1992;4:129–34.
- De Schamphelaere KAC, Janssen CR. A biotic ligand model predicting acute copper toxicity for *Daphnia magna*: the effects of calcium, magnesium, sodium, potassium, and pH. *Env Sci Technol* 2002;36:48–54.
- De Schamphelaere KAC, Heijerick DG, Janssen CR. Refinement and field validation of a biotic ligand model predicting acute copper toxicity to *Daphnia magna*. *Comp Biochem Physiol C* 2002;133:243–58.
- De Schamphelaere KAC, Van Laer L, Deleebeeck NME, Muyssen BTA, Degryse F, Smolders E, et al. Nickel speciation and ecotoxicity in European natural surface waters: development, refinement and validation of bioavailability models. Report prepared for the Nickel Producers Environmental Research Association (NiPERA), Durham, NC, USA. Gent, Belgium: Gent University, Laboratory of Environmental Toxicology and Aquatic Ecology; 2006a.
- De Schamphelaere KAC, Heijerick D, Janssen C. Cross-phylum comparison of a chronic biotic ligand model to predict chronic toxicity of copper to a freshwater rotifer, *Brachionus calyciflorus* (Pallas). *Ecotoxicol Environ Saf* 2006b;63:189–95.
- Deleebeeck NME, De Schamphelaere KAC, Janssen CR. A bioavailability model predicting the toxicity of nickel to rainbow trout (*Oncorhynchus mykiss*) and fathead minnow (*Pimephales promelas*) in synthetic and natural waters. *Ecotoxicol Environ Saf* 2007a;67:1–13.
- Deleebeeck NME, Muyssen BTA, De Laender F, Janssen CR, De Schamphelaere KAC. Comparison of nickel toxicity to cladocerans in soft versus hard surface waters. *Aquat Toxicol* 2007b;84:223–35.
- Deleebeeck NME, De Schamphelaere KAC, Janssen CR. A novel method for predicting chronic nickel bioavailability and toxicity to *Daphnia magna* in artificial and natural waters. *Environ Toxicol Chem* 2008;27:2097–107.
- Deleebeeck NME, De Schamphelaere KAC, Janssen CR. Effects of Mg²⁺ and H⁺ on the toxicity of Ni²⁺ to the unicellular green alga *Pseudokirchneriella subcapitata*: model development and validation with surface waters. *Sci Tot Environ* 2009a;407:1901–14.
- Deleebeeck NME, De Laender F, Chepurnov V, Vyverman W, Janssen CR, De Schamphelaere KAC. A single bioavailability model can accurately predict Ni toxicity to green microalgae in soft and hard surface waters. *Water Res* 2009b;43:1935–47.
- Di Toro DM, Allen HE, Bergman HL, Meyer JS, Paquin PR, Santore RC. Biotic ligand model of the acute toxicity of metals. I. Technical basis. *Environ Toxicol Chem* 2001;20:2378–96.
- Environment Canada. Biological test method: test for measuring the inhibition of growth using the freshwater macrophyte, *Lemna minor*. EPS 1/RM/37; 1999. Ottawa, ON, Canada.
- European Chemicals Bureau. ECB newsletter. Institute for Health and Consumer Protection, Joint Research Centre, European Commission; 2003. Issue 3.
- European Chemicals Bureau. European Union Risk Assessment Report: nickel. Ispra, Italy: European Commission, Joint Research Centre, European Chemicals Bureau; 2008. http://ecb.jrc.ec.europa.eu/DOCUMENTS/Existing-Chemicals/RISK_ASSESSMENT/REPORT/nickelreport311.pdf.
- Lock K, Van Eeckhout H, De Schamphelaere KAC, Criel P, Janssen CR. Development of a biotic ligand model (BLM) predicting nickel toxicity to barley (*Hordeum vulgare*). *Chemosphere* 2007;66:1346–52.
- Mount DR, Mount DI. A simple method of pH control for static and static-renewal aquatic toxicity tests. *Environ Toxicol Chem* 1992;11:609–14.
- Nebeker AV, Savonen C, Baker RJ, McCrady JK. Effects of copper, nickel and zinc on the life cycle of the caddisfly *Clistoronia magnifica* (Limnephilidae). *Environ Toxicol Chem* 1984;3:645–9.
- Nebeker AV, Stinchfield A, Savonen C, Chapman GA. Effects of copper, nickel and zinc on three species of Oregon freshwater snails. *Environ Toxicol Chem* 1986;5:807–11.
- Organization of Economic Co-operation and Development. July Guidelines for the testing of chemicals; revised proposal for a new guideline 221 – *Lemna* sp. growth inhibition test; 2002. Draft Guideline 221.
- Organization of Economic Co-operation and Development. OECD guidelines for testing of chemicals: sediment–water chironomid toxicity test using spiked water; 2006. Method 219.
- Powlesland C, George IC. Acute and chronic toxicity of nickel to larvae of *Chironomus riparius* (Meigen). *Environ Pollut (Ser A)* 1986;42:47–64.
- Ryan AC, Tomasso JR, Klaine SJ. Influence of pH, hardness, dissolved organic carbon concentration and dissolved organic matter source on the acute toxicity of copper to *Daphnia magna* in soft waters: implications for the Biotic Ligand Model. *Environ Toxicol Chem* 2009;28:1663–70.
- Santiago-Fandino VJR. The effects of nickel and cadmium on the growth rate of *Hydra littoralis* and an assessment of the rate of uptake of ⁶³Ni and ¹⁴C by the same organism. *Water Res* 1983;17:917–23.
- Santore RC, Di Toro DM, Paquin PR, Allen HE, Meyer JS. Biotic ligand model of the acute toxicity of metals. II. Application to acute copper toxicity in freshwater fish and *Daphnia*. *Environ Toxicol Chem* 2001;20:2397–402.
- Slooff W, Canton JH. Comparison of the susceptibility of 11 freshwater species to 8 chemical compounds. II. (semi)Chronic toxicity tests. *Aquat Toxicol* 1983;4:277–82.
- Snell TW, Moffat BD. A 2-d life cycle test with the rotifer *Brachionus calyciflorus*. *Environ Toxicol Chem* 1992;11:1249–57.
- Tipping E. Humic ion-binding model VI: an improved description of ion-binding by humic substances. *Aquatic Geochem* 1998;4:3–48.
- U.S. EPA. Aquatic life ambient freshwater quality criteria – copper. Washington, D.C: Office of Water, U.S. Environmental Protection Agency; 2007. EPA-822-R-07-001.
- Van Laer L, Smolders E, Degryse F, De Schamphelaere KAC, Janssen CR. Speciation of nickel in surface waters measured with the Donnan membrane technique. *Anal Chem Acta* 2006;578:195–202.
- Van Sprang PA, Verdonck FAM, Van Assche F, Regoli L, De Schamphelaere KAC. Environmental risk assessment of zinc in European freshwaters: a critical appraisal. *Sci Tot Environ* 2009;407:5373–91.
- Worms IAM, Wilkinson KJ. Ni uptake by a green alga. 2. Validation of equilibrium models for competition effects. *Environ Sci Technol* 2007;41:4264–70.